

Quote

"You're about to cross the finish line, the toughest part is behind you. Just a little more, and you'll have the full power of Machine Learning in your hands."

What We Will Practice In This Chapter

- ❑ Model Tuning
- ❑ Ensemble Learning
- ❑ Unsupervised Learning
 - K-Means Clustering
 - Elbow Method
 - Hierarchical Clustering
 - PCA (Dimensionality Reduction Basics)
- ❑ Hands-On Project
- ❑ A Message

Model Tuning

Model Tuning (Hyperparameter Tuning) is the process of finding the best settings (hyperparameters) for a machine learning model so that it performs better on unseen data.

Why Do We Need Model Tuning?

The first version of a model is rarely the best version.

A model may:

- Underfit
- Overfit
- Give poor performance

Model tuning helps us improve the model and make it more generalized.

Hyperparameter

A Hyperparameter is a parameter whose value is set before training the model.

These parameters control how the learning process works.

Examples

KNN

- `n_neighbors = 3, 5, 7`

Decision Tree

- `max_depth`
- `min_samples_split`
- `max_features`

SVM

- `kernel`
- `C value`

Random Forest

- `n_estimators`

- max_depth
- max_features

Ridge / Lasso

- alpha (regularization strength)

Goal of Hyperparameter Tuning

1. Find the best combination of hyperparameters
2. Improve performance on unseen data
3. Reduce overfitting and underfitting

Cross Validation

Cross Validation is a technique used to evaluate how well a model will perform on unseen data.

Why Use It?

A single train-test split may give misleading results.

Cross validation gives a more reliable estimate of model performance.

K-Fold Cross Validation

In K-Fold Cross Validation, the dataset is divided into K equal parts (folds).

Example:

$k = 5$

The model is trained on 4 folds and tested on the remaining fold.

This process repeats until every fold has been used as a test set once.

The final score is the average of all K results.

Benefits

- 1) Better evaluation
- 2) Less bias
- 3) More reliable performance estimate

Hyperparameter Tuning Methods

Manual Search

Manually changing hyperparameters and checking model performance.

Advantages

- Easy to understand

Disadvantages

- Time consuming
- Not practical for many parameters

Grid Search CV

Grid Search tries every possible combination of specified hyperparameters and uses Cross Validation to find the best one.

Workflow

Data → Model → Cross Validation → Grid Search → Best Model

Advantage

- Finds optimal combination from given grid

Disadvantage

- Computationally expensive

Randomized Search CV

Randomized Search randomly selects combinations of hyperparameters instead of checking every possible combination.

Advantage

- Faster than Grid Search
- Works well for large search spaces

When To Use?

When parameters become too large.

Example:

XGBoost with thousands of combinations.

Difference

Grid Search:

- Checks all combinations

Randomized Search:

- Checks random combinations

Ensemble Learning

Ensemble Learning is a technique where multiple machine learning models are combined to produce a stronger and more accurate prediction.

Main Idea

Instead of trusting one model,
we trust a group of models.

Like hearing the opinion of a crowd.

Types of Ensemble Learning

1. Bagging

2. Boosting

3. Stacking

Bagging (Bootstrap Aggregation)

Bootstrap means randomly sampling data from the original dataset with replacement.

Some data points may appear multiple times.

Purpose

Reduce:

- . Variance
- . Overfitting

Key Idea

Same algorithm

Different random datasets

Different predictions

Final result through voting or averaging.

Random Forest Classifier

Random Forest is an ensemble algorithm that builds multiple Decision Trees and combines their predictions.

Key Characteristics

- Uses bootstrap sampling
- Uses random feature selection
- Uses majority voting

Why Random Forest Works?

Each tree learns slightly differently.

Combining all trees creates a more stable and generalized model.

Final Prediction

Classification:

Majority Voting

Regression:

Average Prediction

Boosting

Boosting is an ensemble technique where models are trained sequentially, and each new model focuses on correcting the mistakes of previous models.

Goal

Convert:

Weak Learners



Strong Learner

Key Idea

Model 1 makes mistakes



Model 2 learns from those mistakes



Model 3 improves further



Continue until strong model is created

Purpose

Reduce:

- . Bias
- . Underfitting

Weak Learner

A model that performs only slightly better than random guessing.

Example:

A shallow Decision Tree.

Popular Boosting Algorithms

AdaBoost

Adaptive Boosting adjusts weights of incorrectly classified samples and focuses more on difficult examples.

Interview Point

One of the earliest boosting algorithms.

Gradient Boosting

Builds models sequentially by minimizing previous errors using gradients.

XGBoost

Extreme Gradient Boosting.

An optimized version of Gradient Boosting.

Why Popular?

- Fast
- Accurate

- Handles large datasets
- Frequently used in competitions and industry

Stacking

Stacking combines predictions from multiple models and feeds them into another model called a Meta Model.

Components

Base Models

Examples:

- SVM
- Logistic Regression
- Decision Tree

These make initial predictions.

Meta Model

Example:

. KNN

Receives predictions from base models and makes final prediction.

Goal

Learn how to combine strengths of multiple algorithms.

Ensemble Learning Summary

Method

Main Idea

Bagging

Parallel learning

Random Forest Many Decision Trees

Boosting

Sequential learning

Stacking

Models on top of models

Important Viva Question

Bagging vs Boosting

Bagging

Boosting

Bagging

Boosting

Parallel training

Sequential training

Reduces variance

Reduces bias

Prevents overfitting Improves weak learners

Random Forest

XGBoost